

Expressive Text-to-Speech Evaluation: Tacotron vs XTTS v2

Author: s4194330 | Date: 6 December 2025 | Runtime: CPU-only conda env `api`

This technical note benchmarks the Tacotron2-DDC baseline against the 2025 Coqui XTTS v2 model for expressive, multilingual text-to-speech. The workshop evaluation text was synthesised using both engines, and the XTTS model generated neutral, happy, and sad variants for an emotion study. Objective signal statistics and runtime profiling underpin qualitative assessments of naturalness, intelligibility, efficiency, and expressive control.

Tacotron2 established end-to-end neural TTS with high intelligibility and smooth prosody [1]. Recent architectures such as XTTS v2 augment multi-speaker and cross-lingual support plus promptable expressiveness via GPT-style acoustic conditioning [2]. This study retains Tacotron2 as the baseline (via the Coqui TTS toolbox) and evaluates XTTS v2 as the recent expressive engine. Both systems ran with PyTorch 2.6 on CPU, so latency reflects non-accelerated inference.

Tacotron2-DDC was invoked through `TTS.api` with default vocoder. XTTS v2 used the released Coqui checkpoint with speaker "Claribel Dervla" and English language setting. Happy and sad variants passed the model's emotion flag; when unavailable, the script falls back to prompt engineering. Inference time was captured with `time.perf_counter`, and audio diagnostics—duration, RMS energy, spectral centroid, and pitch variability—used `librosa`. Results were saved as `tacotron.wav`, `xtts_v2_natural.wav`, `xtts_v2_happy.wav`, and `xtts_v2_sad.wav`.

Results

	System	Inference Time (s)	RTF	Audio Duration (s)	Pitch Std (Hz)	Mean RMS
Tacotron	2-DDC baseline	47.0	1.21	39.0	1113	0.082
	XTTS v2 default neu	172.0	5.27	32.6	1187	0.131
	XTTS v2 emotion=Ha	165.5	5.02	33.0	1183	0.113

XTTS v2 raises pitch variability by +6.7% and mean energy by +60.3% relative to Tacotron, aligning with improved expressiveness. Emotion prompts shift the mean RMS by +67.3% between happy and sad variants, demonstrating controllable dynamics. However, XTTS runs with an RTF of 5.27 on CPU—around 4.4x slower than Tacotron's 1.21—highlighting a latency trade-off.

Qualitatively, both engines produce intelligible speech. Tacotron maintains stable pacing with a neutral tone, yet conveys limited emotional nuance. XTTS v2 introduces noticeable variations in energy and pitch, making the happy rendition brighter and the sad version more subdued. The cost is heavier compute and model downloads (~1.9 GB). For deployment, batching or GPU acceleration is recommended. Voice cloning was not attempted due to the workshop time budget but XTTS provides hooks via `speaker_wav`.

XTTS v2 satisfies the workshop goal of expressive, controllable TTS with measurable gains in prosodic variance, albeit with a significant latency penalty on CPU. Tacotron remains a solid baseline for fast synthesis or limited compute settings. Future work: exploit GPU inference, test voice cloning with few-shot samples, and explore streaming support for low-latency applications.

References

- [1] Y. Wang et al., "Tacotron: Towards End-to-End Speech Synthesis," arXiv:1703.10135, 2017.
- [2] Boson AI, "Higgs Audio V2" (XTTS lineage), <https://github.com/boson-ai/higgs-audio>, 2025.